

Community→Facility

[serviceRequest](#) from community to shared health record

Attribute	Description	Data type or possible values (examples)
Status		
Request intent	order	
Category	Service point	e.g laboratory, imaging, counseling, education, surgical procedures
Priority	For request queueing	High, medium
Code		
Subject	UPI	
Occurrence	Datetime	
Requester	CHV ID	
Performer	Linked facility (mfl code)	
Reason code	nhdd findings or procedure	
Reason reference		
notes	Free Text	
supportingInfo	Additional clinical information	This includes pretreatment medication
patientInstructions		

Sample Payload

Notes

For the requester attribute

1. Use practitioner role where practitioner references CHV ID and organization references CU code. (KMCHUL is available to the FHIR resource)
2. For the referral request, include the same details above for matching with CHVs available in the system.

Questions

1. What is the source of the CHV ID
2. How do we know that the client exists in the community, eCHIS
 - a. For referrals from the facility, we need to know the CU and the CHA will handle the request if the client does not exist yet in eCHIS
 - b. For existing clients, we only need the UPI

Notes

- eHealth bill from Mar 27, 2023
- eCHIS to comply to NHDD
- Trainers to be selected from the team around
 - Korir, Wanyungu and Lesley to lead

Technical Brainstorming

Purpose

To document thoughts and ideas on how to achieve pushing data from CHT to OpenHIM and from OpenHIM to SHR.

Summary

The envisioned connection from eCHIS to the Shared Health Record leverages OpenHIM to mediate payload forwarding. This means that instead of writing directly to a FHIR resource from eCHIS, OpenHIM will do that. The advantage of this is that we can format the payload via mediator logic and monitor the performance of API endpoint calls for any necessary action such as retrying forwarding a message. Our work here therefore, is to write an [outbound push module](#) to relay required data to the mediator, and author the mediator to handle the formatting and forwarding logic. Note that OpenHIM and CHT form eCHIS.

The workstream focuses on two things:

Uploading referral service request data, including screening, medication and referral facility details to the mediator

Formatting the data once it gets to the mediator, into a FHIR resource to update the SHR

Steps:

- Confirm ad map required data points captured in eCHIS
- Write an outbound module to map the data points into a payload to be uploaded to OpenHIM console
- Get access to an OpenHIM endpoint to test the upload

- Build an OpenHIM mediator to format the payload into a FHIR ServiceRequest
- Verify that the ServiceRequest is created successfully

Flow for eCHIS → OpenHIM Data Upload:

- Enable the **mark_for_outbound** transition
- Configure outbound push with the following properties: **relevant_to**, **destination** and **mapping**
- User submits service records in the Android app
- App syncs to the eCHIS
- eCHIS pushes relevant data to OpenHIM

Flow for OpenHIM → FHIR Server Service Request Upload:

- OpenHIM mediator gets published
- Mediator receives and validates the payload from eCHIS
- Mediator queries KMFL to populate the performer object of the ServiceRequest
 - Not required if we already have the HF code in eCHIS
- Mediator queries KMCHUL to populate the requester object of the ServiceRequest
 - Not required if we already have the CU code in eCHIS
- Mediator queries NHDD to populate the supportingInfo object of the ServiceRequest
 - Not required if we already have the Concept code in eCHIS
- Mediator formats the payload into a proper JSON
- Mediator forwards the payload to the ServiceRequest endpoint
- Transaction is marked successful

Payload Mapping

Outbound Module

Service forms data points mapping

Service	Screening	Medication	Other supporting info	upi
Under 5 assessment (0 to 2 months)	cannot_feed convulsed_or_fitted breath_count chest_indrawing has_fever child_temp no_move has_yellow_soles bleeding_umbilical_stump local_infection_signs birth_weight_chart is_breastfeeding_exclusive ly	--no danger signs r_has_taken_amox r_given_amox r_given_ors r_given_zinc r_given_al r_given_paracetamol r_dewormed_child --has danger signs r_dt_taken_amox r_dt_given_amox r_dt_given_ors r_dt_diarrhoea_given_zinc r_dt_fever_given_paracetamol		
Under 5 assessment (2 months to 5 years)	has_cough has_contact_with_tb breath_count u5_has_fast_breathing chest_indrawing has_diarrhoea diarrhoea_duration blood_in_stool has_fever fever_duration			

	child_temp has_convulsions has_difficulty_drinking_feeding cannot_drink_feed is_vomiting is_vomiting_everything is_sleepy_unconscious has_taken_alo2_month_has_danger_signs malaria_test_result repeat_malaria_test_result muac_color has_swollen_feet non_treatable_condition still_breastfeeding breastfed_in_last_24hrs has_taken_food type_of_food_taken			
Immunization	--from u5 assessment refer_for_vaccines vit_a_received deworming_upto_date is_participating_growth_monitoring delayed_development_milestones --from immunization form vaccines_for_referral vitamin_a_doses_given has_deworming_status_uptodate is_participating_in_monthly_growth_monitoring has_signs_of_delayed_milestones	Not applicable		

Pregnancy home visit (ANC)	1. Upi 2. name 3. has_lost_pregnancy_within_last_month 4. danger_signs 5. other_danger_sign 6. takes_iron_or_folate_supplements 7. mental_danger_signs 8. other_mental_danger_sign 9. observed_mental_danger_signs 10. other_observed_mental_danger_signs	Not applicable	1. is_still_pregnant 2. has_started ANC 3. ANC_upto_date 4. has_lost_pregnancy_within_last_month_subsequent_visit 5. pregnancy_test_result 6. is_pregnant 7. is_on_family_planning 8. period_of_pregnancy 9. known_lmp 10. next ANC_visit_date	
Postnatal care (mother)	mother_danger_signs mother_danger_signs_other mental_signs observed_mental_signs	Not applicable	has_pnc_up_to_date has_started_family_planning next_pnc_visit_date next_appointment_date_for_family_planning_service	
Postnatal care (newborn)	newborn_danger_signs breaths_per_minute temperature has_updated_immunization_status --get the same set of details above from the contact document created off the Postnatal care (mother) service form	Not applicable	newborn_education caregiver_educated_on_newborn_care	
Family planning	interested_in_fp	Not applicable	fp_method	

	has_side_effects		refilled_today (date) num_pieces num_cycles next_appointment_date	
Over 5 assessment	symptoms symptoms_other has_fever fever_duration tb_symptoms has_diarrhea diarrhea_duration has_blood_in_stool is_diabetic diabetes_symptoms is_hypertensive hypertension_symptoms systolic_blood_pressure diastolic_blood_pressure mental_signs observed_mental_signs	given_al given_paracetamol	has_taken_al rdt_result repeat_rdt_result is_on_tb_treatment knows_hiv_status is_on_diabetes_medication is_on_hypertension_medication is_screened_cancer has_observed_signs_of_violence	
Referral follow up	needs_to_go_to_health_facility	Not applicable		
Defaulter follow up	defaulted_services other_defaulted_services attended_visit needs_to_go_to_health_facility	Not applicable	service_provider service_provider_other why_default	
Treatment follow up	progress side_effects	Not applicable	is_taking_treatment_as_prescribed	

Payload Structure to OpenHIM

```
{  
  upi: '<e.g. MOHQW3RTY54>',  
  service: '<e.g. family planning>',  
  fields: '<json report fields object, include the NHDD codes>' Is pushing a JSON  
object raw even possible?  
}
```

- UPI shall be sourced from the contact document, after the attribute has been added via the community → client registry workflow.
- Service is more or less similar to the form attribute
- Fields implies the intention to do minimal processing and leaving a bulk of the logic to the OpenHIM mediator.
 - This is because the data points are organized in different groups in the JSON doc and extracting them one after the other might not be ideal.
 - The downside is that some reports are long, e.g. assessment, and might be worth exploring whether there are limits to the number of key value pairs that can be posted via Outbound push.

Notes

1. Link facility

- Each CHV Area JSON document has the link facility code and name as attributes e.g.

```
{  
  ...  
  "link_facility_code": "13778",  
  "link_facility_name": "Mariwa Health Center",
```

...

}

This can be added to the payload.

- Link facilities codes are obtained from the [KMFL](#), see also [readthedocs](#).

2. CHV code from CHV registry

- There is an existing CHV registry, though not updated real time, available on the [KMFL](#).
- For this to be usable, we need to update eCHIS with the CHV code after the registry has been updated to include all CHVs offering community services. The update is usually done at the sub county level and can be cascaded to CHAs to hasten the process.
- Once available, the code can be added to the payload.

3. CU code

- Community Health Units are available on the [KMCHUL](#).
- When updated, eCHIS will be updated to include the CU code attribute.
- Similar to the two above, the updates are done at sub county level.

Questions

1. How soon can the above be updated?

- With the current deployment in Migori, the update can take two weeks, to be jointly done by the County team and Lwala.
- Counties intending to roll out in the near future shall be required to update their details in the various registries. This will be part of the process for gathering details for users and contacts creation.

Mediator to produce Service Request

[Documentation](#)